1. Hashing a Password (SHA-256)

```
import hashlib
password = "mypassword123"
hashed_password = hashlib.sha256(password.encode()).hexdigest()
print(hashed_password)
```

```
6e659deaa85842cdabb5c6305fcc40033ba43772ec00d45c2a3c921741a5e377
```

2. Faking (Generating Fake Data)

```
pip install faker
```

```
Collecting faker
  Downloading faker-37.1.0-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: tzdata in /usr/local/lib/python3.11/dist-packages (from faker) (2025.2)
Downloading faker-37.1.0-py3-none-any.whl (1.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.9/1.9 MB 17.8 MB/s eta 0:00:00
Installing collected packages: faker
Successfully installed faker-37.1.0
```

```
from faker import Faker
fake = Faker()
print(fake.name())
print(fake.email())
print(fake.address())
```

```
Sarah Franklin
jacobsmith@example.net
66005 Oconnor Greens
New Adamstad, VI 49351
```

3. Masking Sensitive Data (Credit Card)

```
def mask_credit_card(card_number):
    return "*" * (len(card_number)-4)+card_number[-4:]
card_number = "1234567890123456"
print(masked_card)
```

```
************3456
```

4. Data Anonymization (Drop name column)

```
import pandas as pd
data = {'name': ['Alice', 'Bob', 'Charlie'], 'age': [25, 30, 35], "salary":[50000, 60000, 70000]}
df = pd.DataFrame(data)
df = df.drop('name', axis=1)

print(df)
```

```
   age  salary
0   25   50000
1   30   60000
2   35   70000
```

5. Pseudonymization (Replace name with User ID)

```
user_info = {"id": 101, "name": "Alice johnson"}
user_info["Name"]="user"+ str(user_info["id"])
print(user_info)
```

```
{'id': 101, 'name': 'Alice johnson', 'Name': 'user101'}
```